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Executive summary

The coordinated anticipatory action (AA) framework for cyclones in Bangladesh, initiated in January 2023 and revised in March 2025, leverages early warning systems and pre-arranged financing to mitigate humanitarian impacts of cyclones in the Bay of Bengal. Building on extensive experience, including past AA efforts during monsoon floods, the framework targets interventions before predicted hazards occur. Led by the Resident Coordinator (RC), the framework is integrated with existing coordination mechanisms and aims to enhance response efficiency while protecting development gains. By targeting at-risk populations and coordinating efforts among stakeholders, this framework strengthens resilience and reduces the impact of cyclones on vulnerable communities in Bangladesh. A total of \$8,026,987 (\$4,000,000 from the UN Central Emergency Response Fund/CERF and \$4,026,987 from BDRCS¹, BRAC, STEP, UNFPA, UNICEF, and WFP) is pre-arranged under this framework and will be disbursed for AA interventions should pre-agreed trigger thresholds specified in this framework be reached.

The framework uses scientific information on forecasts and triggers to automatically enable the implementation of a pre-agreed action plan in the target districts. A learning agenda is also part of the framework to extract key learning and enable a continuous evolution of the framework in the future.

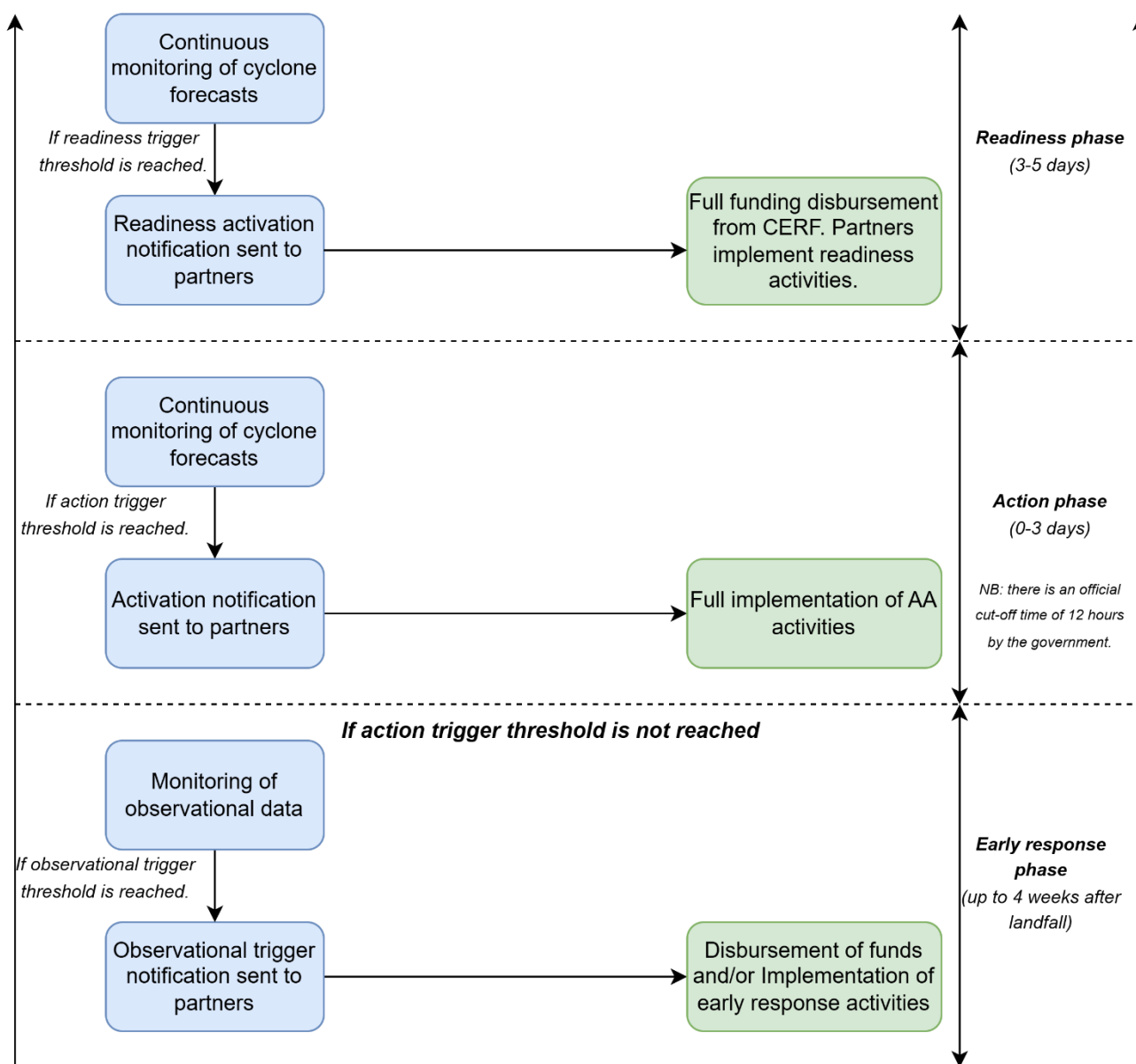
Current structure of the framework and activation protocol

The overall structure of the framework is summarised as follows:

- During the cyclone seasons (April to June and October to November), cyclone forecasts are continuously monitored by the monitoring team for a potential activation of the framework.
- If the readiness criteria are met, the monitoring team sends a standard communication to partners. Readiness activation can happen between 3 and 5 days before landfall.
- Upon readiness activation of the framework, CERF disburses the full amount to partners. However, partners will only be able to implement readiness activities (as specified in the CERF projects) and will have to wait for the activation of the action trigger to start the full implementation of activities as defined in Section 3.
- In case of an activation of the action trigger, the monitoring team sends a further communication to partners.
- Upon action activation of the framework partners can start implementing all activities immediately with no further approval required.
- In case of a missing readiness activation (i.e. an action trigger is reached not preceded by a readiness trigger) the disbursement of funding will happen as soon as the action trigger goes off, when implementation of activities also starts.
- An observational trigger is included to cover scenarios where a cyclone does not meet the readiness or action trigger thresholds ahead of landfall but ends up causing severe humanitarian impacts in the target districts. If the observational trigger is reached, agencies are authorised to spend 100% of the funds. The observational trigger can only be reached if the AA action trigger was not reached ahead of landfall.

¹ BDRCS: Bangladesh Red Crescent Society.

AA activation workflow



1. Introduction

In January 2023, the Under-Secretary-General and Emergency Relief Coordinator (USG/ERC) granted a request by the Bangladesh Resident Coordinator to continue OCHA's support in facilitating and financing coordinated anticipatory action in line with the Humanitarian Coordination Task Team's (HCTT) strategy to increase the geographic reach and to expand AA to other hazards. This enabled the geographical expansion of the anticipatory action framework for monsoon floods in Bangladesh in early 2023 as well as the development of this anticipatory action framework for cyclones.

Extensive anticipatory action experience has been built over the past few years in Bangladesh for a variety of hazards. Regarding monsoon river floods, the Bangladesh Red Crescent Society (BDRCS) pioneered the approach in 2015, developing trigger-thresholds and an Early Action Protocol (EAP). A previous OCHA-coordinated anticipatory action framework for floods triggered in 2020 and 2024 and yielded extensive learning and impact evaluations providing a basis for the improvement of the floods framework ever since.² Furthermore, an Anticipatory Action Technical Working Group exists within the HCTT structure, complementing the Government-led Taskforce on Forecast-based Financing/Action (FbF/A). Both OCHA-facilitated AA frameworks (for floods and cyclones) are also a contribution to the UN Secretary-General's Early Warning for All initiative (EW4All).³

Risk/Hazard

Bangladesh is highly vulnerable to climate-related shocks and stresses, including cyclonic events, which usually occur from April to June and October to November each year. The country's geographical location, with a vast coastline along the Bay of Bengal, makes it particularly vulnerable to tropical cyclones. In recent decades, while the preparedness programme for cyclones by the Government and development partners has improved substantially in saving lives, cyclones still threaten lives and livelihoods and cause significant damage to assets and infrastructure. According to the Ministry of Environment, Forest and Climate Change (MoEF), Bangladesh is hit by a severe cyclone every 3 years on average. According to the Bangladesh Meteorological Department (BMD), 10 cyclones impacted coastal districts of Bangladesh over the past 32 years (1991 to 2023), and six tropical cyclones with wind speeds exceeding 118 km/h formed in the Bay of Bengal.

Anticipatory Action

By using early warning systems and scientific advances in predicting disasters, anticipatory action maximises the window of opportunity between the moment of prediction and the arrival of a specific forecasted shock to trigger interventions that prevent or mitigate imminent humanitarian impacts. This approach leads to more effective, efficient and dignified responses and protects hard-won development gains. Anticipatory actions aim to interrupt the severity of a shock – in this case the cyclone impact – on vulnerable households by targeting populations most at risk. The targeting of anticipatory action ("people at risk") is therefore distinct and different from targeting for the humanitarian response which is determined by existing need ("people in need").

Anticipatory action is defined as acting ahead of predicted hazards or their peak impact to prevent or reduce acute humanitarian impacts before they fully unfold. Building on growing evidence that anticipatory action is more effective than traditional humanitarian response, OCHA has been coordinating and financing collective anticipatory action frameworks for predictable hazards, such as cyclones, floods, droughts and certain disease outbreaks (cholera). Each framework comprises 3 core elements, all of which are underpinned by a clear learning, monitoring and evaluation plan:

² For an entry point to the learning outcomes, please see here: <https://www.unocha.org/our-work/humanitarian-financing/anticipatory-action/summary-bangladesh-pilot>

³ <https://public.wmo.int/en/earlywarningsforall>

- A robust forecasting embedded in a clear decision-making process (the **model**).
- Pre-agreed action plans that can fundamentally alter the trajectory of the crisis (the **delivery**).
- Pre-arranged finance (the **money**).

In line with best practices for anticipatory action, this framework entails a pre-agreed decision-making process based on cyclone forecasting and observation, a pre-agreed action plan by pre-identified actors which are enabled by pre-arranged financing from different funding sources. In addition, documenting evidence and learning is part of the framework.

Prior anticipatory action in Bangladesh

Bangladesh has built extensive anticipatory action expertise thanks to a variety of AA initiatives implemented by humanitarian actors. The Bangladesh Red Crescent Society (BDRCS) and the World Food Programme (WFP) have been pioneers in implementing anticipatory action in Bangladesh, having piloted this approach in 2015, 2017, and 2019, with a particular focus on anticipatory action cash assistance. BDRCS has played a significant role in the implementation of AA cash assistance, while WFP has worked extensively on the institutionalisation of anticipatory action within the region. The BDRCS and WFP developed trigger thresholds and an Early Action Protocol (EAP) for Monsoon Floods which was adopted by the OCHA-facilitated and CERF-supported AA Framework and activated during the 2020 and 2024 monsoon flood seasons.

In 2019, an “Anticipatory Action Technical Working Group” was established, bringing together interested agencies under the leadership of BDRCS. The group was officially launched under the auspices of the Ministry of Disaster Management and Relief (MoDMR) and is linked to the HCTT co-chaired by the UN Resident Coordinator (RC) and MoDMR Senior Secretary.

In 2020, OCHA facilitated a collective approach to anticipatory action as part of a global pilot phase. Building on existing structures and experiences, the CERF-supported OCHA pilot framework pre-established when and on what basis financing and action would be triggered ahead of a specific monsoon flooding peak; how much funding would go to which agency; and what activities the funding would be used for and in what time frame. On 4 July 2020, severe floods were forecasted from 18 July onward. The framework was triggered, and CERF funding was released within minutes. By the time the water reached life-threatening levels, more than 220,000 people had already received assistance through WFP, FAO and UNFPA, which worked with the Red Cross/Red Crescent, 10 local non-governmental organisations (NGOs) and the Government. WFP and some NGOs also carried out similar, independent anticipatory actions during the same time. Using the IFRC DREF, BDRCS carried out the anticipatory action for the 2020 monsoon floods on 4 July 2020, before the first peak of the flood under FbA.

In 2021, the revised AA framework for monsoon floods was endorsed by the RC and pre-approved by the Emergency Relief Coordinator (ERC), for multi-sectoral interventions carried out by the UN and the Red Cross/Red Crescent in close collaboration with NGOs and the Government through CERF funding. The objective of the revised pilot was to further scale up the quality and quantity of collective anticipatory humanitarian action to people at risk of predicted severe monsoon flooding of the Jamuna River in Bangladesh, covering five highly vulnerable districts (Bogura (Bogra), Gaibandha, Kurigram, Jamalpur, and Sirajganj) and using available forecasts with a two-step trigger system to predict severe monsoon floods. Based on this experience, the anticipatory action framework for monsoon in Bangladesh was expanded in 2023, to increase the reach in the Jamuna River basin and add coverage of the Padma River basin.

In July 2024, the collective floods framework prompting the immediate release of \$6.2 million from CERF with partners contributing another \$2.3 million. Over 80 % of the assistance was delivered within 5 days of the activation, reaching communities ahead of peak flooding and before they were cut off. Together, the UN, the Bangladesh Red Crescent Society, the Government, and other partners reached over 500,000 people prior to

severe monsoon flooding in the Jamuna River basin. Affected people received multi-purpose cash transfers, water purification supplies, agricultural support, dignity and baby kits, and information services on gender-based violence. With this anticipatory and fast support, affected community members secured for example two weeks' worth of food, stored their valuables in safe places and bought a portable stove. As the Jamuna River swelled above the danger line, flooding entire villages, this early aid was even more crucial, given that stagnant floodwaters cut off people's access to post-disaster relief for weeks after the peak floods. Learning from this activation showed the effectiveness of the approach.

Overarching framework and core principles

The Resident Coordinator (RC) is responsible for the overall coordination and accountability of international assistance by the United Nations in the country, which includes CERF allocations. Under the leadership of the RC, the pilot is anchored in existing in-country humanitarian coordination mechanisms and modalities for anticipatory action, which is also referred to as forecast-based action (FbA) or forecast-based financing (FbF).

The framework builds on existing protocols to reinforce existing coordination mechanisms, including the BDRCS and WFP Co-led Anticipatory Action Technical Working Group, the Needs Assessment Working Group (NAWG), the Inter-Cluster Working Group (ICCG), Humanitarian Advisory Group (HAG) and the HCTT. The framework is also in line with existing tools, including the relevant Joint Contingency Plan framework for climate-related disasters in the context of the HCTT Nexus Strategy 2021-25.

2. The Model: Forecast and Triggers

Forecast

Monitoring will take place during cyclone season in Bangladesh, which occurs twice per year from April to June and October to November. The AA-TWG will consolidate the forecast and communicate it to the HCTT members via email. The sources used for the forecast are the Bangladesh Meteorological Department (BMD), India Meteorological Department (IMD), European Centre for Medium-Range Weather Forecasts (ECMWF), Global Forecast System (GFS), and Joint Typhoon Warning Centre (JTWC). The ECMWF forecast also shows the likelihood of landfall and potentially impacted areas which is useful for activation decisions.

The trigger system has been designed to be aligned and make use of the early warning systems of the BMD and to other existing projects such as the Bangladesh Red Crescent Society (BDRCS) Early Action Protocol.

Trigger

A trigger for anticipatory action should be based on a set of criteria to help answer the questions of when and where to act before an imminent cyclone. It should determine when a cyclone becomes an out-of-the-ordinary (severe) shock that causes high level of humanitarian impact for the exposed vulnerable community. For this framework, a 1-in-4.75-year return period or greater is considered which corresponds to an extreme cyclone event with expected wind speed in the target coastal districts of at least 118 km/h, corresponding to either:

- a 25% household asset damage or higher or
- at least 40% of the population in a union likely to be affected.

In order to increase the available lead time and ensure that partners are ready to implement activities in the short time available ahead of landfall the trigger has been designed as follows:

Stage I: Readiness (release of funding from CERF and start of readiness activities.)

- At least two models from BMD, IMD, ECMWF, GFS and JTWC forecast wind speeds of 118 km/h (also defined in national early action protocol approved by the government) or higher in the coastal pilot districts of Bangladesh with a lead time of 3-5 days.

Stage II: Action (start of implementation of activities by partners)

- Well-formed low pressure or depression has formed in the Bay of Bengal and at least two models from BMD, IMD, ECMWF, GFS and JTWC forecast wind speeds of 118 km/h or higher in the coastal districts of Bangladesh with a lead time of up to 72 hours in the coastal pilot districts.

Stage III: Observational Trigger at Landfall (start early response activities)

- The observed wind speed at the time of landfall is 118 km/h or higher from BMD or IMD or at least two models from ECMWF, GFS, and JTWC confirm wind speeds of 118 km/h or higher in the coastal pilot districts of Bangladesh at landfall or crossover. The observational trigger will only be activated if the action trigger fails to activate and if the observational trigger thresholds are reached.

Handling forecast-based edge-cases

The framework will activate based on forecasts whenever thresholds are reached for the time of landfall. If activation occurs with less than 24 hours' lead time ahead of landfall, agencies that can safely implement activities may proceed, while those unable to do so may shift to early response. This should be decided by the Resident Coordinator in consultation with the participating agencies and the Humanitarian Coordination Task Team. Agencies should only implement activities if it is safe to do so and therefore should not jeopardize the security and safety of targeted people, implementing partners and their personnel. Agencies should also adhere to all local regulations with regards to the closure of facilities and cessation of activities less than 12 hours before landfall. Additionally, it is to be noted that there is a 50% chance that the readiness trigger is not reached while the action trigger is, and vice versa.

Early Response

Once the observational trigger threshold is reached, there is a possibility that the targeted districts for AA could be different from the districts that are actually impacted by the cyclone. In such cases, a geographical reprioritization should be conducted to assess the districts that are most affected and where the needs are highest. In terms of activities to be implemented, the AA activities under this framework are largely dual-use and therefore mostly (with the exception of a few) also eligible for immediate post-shock response and should be implemented as such as per the CERF project documents. Additional and significant changes that are necessary should be made to CERF projects through a standard CERF project revision request. CERF will handle such requests with the appropriate flexibility.

Monitoring steps

The monitoring team, composed of Ahmadul Hassan (AA trigger expert), Pauline Ndirangu (Data Scientist, OCHA) and Leonardo Milano (Data Science Team Lead, OCHA), is responsible for tracking forecasts to alert partners in case of an activation. The analysis of the forecasts is composed of the following steps:

- Based on the expected location of landfall the distance to the target unions is calculated.
- A wind speed reduction factor (developed under the DREF EAP cyclone of IFRC) is applied to estimate the expected wind speed velocity for each union.
- Unions/upazilas exceeding the activation thresholds are identified.

- Trigger criteria are verified based on the trigger statement above
- Situation report is produced and provided to OCHA AA partners.

Roles and responsibilities

Who	What	How	When
AA Working Group - Cyclone focal point/Monitoring team	BMD /IMD / ECMWF / GFS / JTWC forecast analysis	Populates (twice a day) the forecast and observation in the log sheet	As soon as forecast is produced, and before forecast is truncated to 72 hours
AA Working Group - Cyclone focal point/Monitoring team	Evaluate whether trigger has been met, and send email to distribution list	Communicates status of 5-day readiness and/or 72 action and/or observational trigger to the HCTT members, RC/RCO as well as concerned UN Agencies, OCHA and other partners through email. A WhatsApp Group is also used to facilitate communication. The communication includes an intervention map.	
CERF Secretariat	When a readiness trigger is reached (or an action/observational trigger without a previous readiness trigger), CERF will immediately send the approval letters to the agencies	Endorsed framework and CERF application by RC and Emergency Relief Coordinator (ERC) Pre-approved CERF approval letters to UN agencies	As triggers are reached
RC	The RC can decide to pause the activation process upon discussion with the MoDMR	Direct contact between RC and MoDRM	Once the framework is activated, in the possible outlier cases where a track of a storm clearly would result in minimal impact.
MoDMR	The MoDMR can discuss with the RC if there is a need to pause the activation.	Direct contact between RC and MoDMR (it can be facilitated by OCHA HoO)	Once the framework is activated, in the possible outlier cases where a track of a storm clearly would result in minimal impact.

Any triggering of the framework should lead to action. However, for theoretically possible outlier cases where a track of a storm clearly would result in minimal impact, the Resident Coordinator, in consultation with the government, may pause any activation.

Challenges and potential improvements in the future

- Need for impact-based forecasting after triggers are reached. This could start and be improved continuously.

- Lead time for action may be extremely short. More investment in the science of cyclone forecasting could increase lead times and accuracy.
- Tropical cyclones through the lens of wind speed are only one hazard resulting from storms, alongside, e.g., rainfall and storm surge. Bangladesh faces other predictable hazards that are not included in this anticipatory action framework, such as monsoon floods, and the anticipatory action framework for floods could potentially be integrated into the cyclone anticipatory action framework in the future.

3. The Delivery: Action Plan

Overview

Lack of resilience, early warning, and resources before a cyclone impact may cause poor households in affected areas to lose their lives, houses, assets, (including agricultural assets), crops, food grains, and income from livestock. Affected people often lose their agricultural assets like seeds, grains and may also be forced to adopt negative coping strategies such as skipping meals, reducing portion sizes and consuming lower quality food as a consequence. Women and girls often struggle to meet essential material needs during a crisis, leading to compromises on health, creating limited mobility and isolation, in addition to a greater risk of gender-based violence. Evidence from past emergencies demonstrates that the menstrual health of adolescent girls worsens during crises. High mortality of livestock was observed during cyclones due to lack of shelter as well as damages in the fisheries sector due to flooding. These effects can have long-term impacts on communities, eroding development gains and exacerbating susceptibility to future shocks.

The table below presents the vulnerability levels of Bangladesh's coastal, sea-facing districts. These districts were ranked using a combination of INFORM RISK analysis indicators, including exposure and vulnerability, as well as the frequency of cyclone landfalls along the Bangladeshi coast where wind speeds exceeded 118 km/h (Category I) between 1977 and 2023. Initially, the framework targeted only the top five districts identified as most at risk. However, with the expanded scope of the framework and the increased participation of additional partners and different sources of funds, AA activities may now be implemented across all sea-facing districts. It is important to note, however, that CERF funding is currently limited to districts most exposed: Satkhira, Patuakhali, Bhola, Noakhali, Barguna, and Khulna.

Rank	Districts	INFORM Natural Hazard	INFORM Vulnerability	Frequency Exceeding Wind Speed of 118 km/h (Cat.1)	Score
1	Satkhira	6.9	5.5	0.1304	4.95
2	Patuakhali	5.6	6.7	0.1304	4.89
3	Bhola	5.0	5.8	0.1522	4.41
4	Noakhali	5.9	5.6	0.1304	4.31
5	Barguna	7.2	6.4	0.0870	4.01
6	Khulna	6.0	5.9	0.0870	3.08
7	Pirojpur	5.4	6.1	0.0870	2.86
8	Bagerhat	6.0	6.5	0.0652	2.54

Anticipatory and early response actions that are effective in preventing or reducing the humanitarian impact of severe cyclones need to be delivered within the window of opportunity, i.e., between the trigger and the

peak impact of the cyclone. Agencies thus need the operational capacity (thematic, logistic, administrative, financial, human resources) to implement the action effectively. For this framework, interventions have been selected to be rooted in the current operating environment, including the agency's capacity to deliver, the ability to mitigate the impact of cyclones, and learning from past experiences. To maximise impact, the anticipatory action interventions of the participating agencies are coordinated and geographically aligned with one another. The details for UN-lead agency activities as part of the framework can be found in the CERF applications – an overview is provided below.

Interventions by the Food and Agriculture Organization (FAO)

Objectives

Protecting livelihood assets through the provision of anticipatory or early actions in cyclone-prone coastal areas.

FAO-led actions

FAO's anticipatory action project aims to protect the livelihoods and agriculture assets of 5,550 farming households in the four identified most vulnerable districts – Satkhira, Barguna, Patuakhali and Noakhali – as well as to reach some 25,000 households with early warning messages. FAO's main activities are:

Readiness phase

- Dissemination of Early Warning Messages (EWM): voice and SMS to 106,500 people (25,000 households)] and via loudspeaker at community level and union/shelter catchment area in coordination with CPP and implementing partners
- Government partner sensitization
- Facilitation of cyclone shelter preparation for cattle management in coordination with Shelter Management Committee, Local Disaster Management Committee and Department of Livestock Services (DLS)
- Community mobilization for safe livestock evacuation
- Procurement of animal feed

Action phase

- If forecast-based trigger is reached (pre-landfall): Dissemination of Early Warning Messages (EWM): voice and SMS to 106,500 people (25,000 households)] and via loudspeaker at community level and union/shelter catchment area in coordination with CPP and implementing partners
- Provision of animal feed for evacuated cattle at the cyclone shelter (60 shelters at selected districts) during the shelter period (approx. 200 cattle/ shelter)
- Facilitation of shelter-based cattle management during evacuation time in coordination with the Department of Livestock Services (DLS)
- Distribution of unconditional cash grants (@USD 41.32 or BDT 5000 through Mobile Money Transfer) to listed 5,550 farming households to take preparation (transportation/ evacuation/animal shelter raising/ repair/ destocking etc.)
- Distribution of Information, Education and Communication (IEC) materials on livestock management in emergencies
- Post-distribution monitoring as part of FAO AAP standard as well as participation in the common independent evaluation

Interventions by the UN Population Fund (UNFPA)

Objectives

Integrated anticipatory or early actions for gender-based violence (GBV) and sexual and reproductive health (SRH), to mitigate cyclone-induced risks to women, girls and third gender persons including pregnant women and persons with disabilities.

UNFPA-led actions

UNFPA's interventions will seek to mitigate the impacts of GBV, maternal mortality, and challenges in the menstrual health of adolescent girls during a cyclone event, by implementing activities in two high-risk districts – Patuakhali and Noakhali. Concretely, UNFPA will target 9,050 at-risk women and adolescent girls, including 2,000 pregnant women and 200 trans women.

The integrated GBV-SRH intervention in the two targeted districts focuses on:

- Distribution of 2,500 dignity kits among women and transgender group in the two districts, along with key lifesaving messages on SRH services, referral and GBV referral and risk mitigation;
- Deployment of 9 MHPSS Counsellors to vulnerable locations to provide mental health and psychosocial support (MHPSS) and lifesaving information to women and girls. (Brothels & Shelters)
- Deployment of Community Women Support Volunteers (CWSVs) in Noakhali and Patuakhali to support safe evacuation of women, girls and other vulnerable groups and promote prevention of sexual exploitation and abuse (PSEA).
- Provision of 2,000 Menstrual Health Management (MHM) kits with two-way transportation cash support for adolescent girls (10 to 19 years) at risk of/or affected by cyclone impacts. Each kit contains age-appropriate menstrual health and hygiene supplies to cover a minimum of three-months menstrual cycles of an adolescent girl.
- Provision of cash assistance to 1,500 adolescent girls (10-19 years) to manage their Menstrual Health and Hygiene (MHH) along with key lifesaving messages on SRH services, referral and GBV referral and risk mitigation.
- Provision of 2,000 Baby kits will to pregnant women in two districts.
- Deployment of midwives at the shelters/union facilities, equipped with essential supplies and logistics. Some midwives will support safe evacuation of pregnant women in union level facilities as per need.
- Provision of cash support to facilitate obstetric care services for 50 pregnant women with complicated pregnancies.
- Dissemination of key lifesaving information on SRHR services and referral, GBV referral at the evacuation centres for all target groups.

Interventions by the UN Children's Fund (UNICEF)

Objectives

Provision of integrated multi-sectoral high-impact anticipatory or early action interventions in WASH, Health, and Multi-Purpose Cash for vulnerable women and children in all five districts identified for this framework – Bhola, Barguna and Patuakhali in Barisal Division; Noakhali in Chattogram Division; and Satkhira in Khulna Division.

UNICEF-led actions

UNICEF's project will target up to 150,000 of the most vulnerable community members affected by the cyclone in the five targeted districts. The integrated interventions will ensure that the affected populations have access to a comprehensive package of services that will reduce the risk of life-threatening water and sanitation-related diseases in affected areas (such as acute watery diarrhoea); provide an opportunity for children to continue their education and strengthen systems and communities' resilience through cash transfers.

The integrated multi-sectoral interventions will provide urgent and quick-impact lifesaving anticipatory or early action interventions, such as:

- Distribution of life-saving WASH supplies, including water purification tablets (WPTs) and Jerrycans.
- Deployment of 20 mobile water treatment plants in the potential evacuation point and cyclone shelters to provide safe drinking water.
- Distribution of hygiene kits including Menstrual hygiene management materials.
- Deployment of medical teams with volunteers and essential medical supplies.
- Movement of vaccines, essential medicines, cold chain and medical equipment to higher floors to prevent storm damage.
- Relocation of pregnant women, newborns, and critically ill patients to safer locations
- Engagement of volunteers to sensitize high risk pregnant women and families with newborn to provide MNCH education in shelters with medical team.
- Cash assistance (5000 BDT/41.32 USD/Per HH) through mobile money transfer, including payment verification and grievance management.

Interventions by the World Food Programme (WFP)

Objectives

Anticipatory or early action to support vulnerable cyclone-affected households with early warning messages, cash and food assistance

WFP-led anticipatory actions

WFP will be focusing its cyclone interventions on socio-economically poor and vulnerable households in the cyclone-prone districts of Khulna, Satkhira, Potuakhali Barguna and Bhola. Forty-five thousand households will be selected and verification of the bKash accounts will be undertaken in a pre-readiness phase, to enable instant cash transfer immediately upon activation of the trigger. WFP will inform beneficiaries of their entitlement through mobile messages (SMS).

Once activated:

- The 35,920 beneficiary households will receive a one-off unconditional cash transfer of BDT5,000 (equivalent to US\$41.25) via mobile cash transfer (bKash) before the cyclone hits. The transfer value of BDT5,000 was calculated to cover 75 per cent of the minimum expenditure basket (MEB) for a household comprising five members.
- WFP will also provide immediate food assistance to those evacuated and sheltered temporarily at cyclone shelters. With the CERF funds, WFP will distribute fortified biscuits to 25,000 households (125,000 people).
- Cyclone early warning messages will also be disseminated, including in local languages and to remote villages. The dissemination of the last-mile early warning messages will be continued jointly with the Cyclone Preparedness Program (CPP) volunteers and within the community.

4. The Money - Pre-arranged Funds

Pre-arranged financing to fuel anticipatory action is critical to enable timely, effective anticipatory action, especially in sudden-onset disasters. To ensure sustainability of anticipatory action approaches, a key recommendation from the Emergency Relief Coordinator (ERC) is for agencies to seek alternative sources of financing for coordinated anticipatory action over the longer-term. This is in addition to the considerable resources invested by partner agencies themselves in building coordinated anticipatory action alongside other preparedness. CERF has provided an envelope of \$ 4,000,000 in support of this framework. This is complemented by an additional \$5,951,987 from BDRCS, BRAC, STEP, UNFPA, UNICEF, Islamic Relief, Save the Children and START NETWORK and WFP. This brings the total pre-arranged amount under this framework to \$9,951,987

Financing overview

Agency	CERF Funding (US\$)	Co-funding (US\$)	Total (US\$)	Remarks
FAO	500,000	-	500,000	
UNFPA	300,000	15,930	315,930	
UNICEF	1,100,000	1,900,000	3,000,000	
WFP	2,100,000	1,600,000	3,700,000	
BDRCS	n/a	441,950	464,047	DREF
		22,097		Internal (quick fund) ⁴
BRAC	n/a	8,500	8,500	Internal fund
STEP	n/a	38,510	38,510	Tentative
Start Network	n/a	1,775,000	1,775,000	Start ready
Islamic Relief	n/a	64,000	64,000 ⁵	
Save the Children	n/a	150,000	150,000	ECHO and GFFO and SC Humanitarian Fund
Total	4,000,000	5,951,987	9,951,987	

Central Emergency Response Fund (CERF)

The ERC agreed to allocate US\$4 million from CERF to support this anticipatory action/early action framework for cyclones in Bangladesh. CERF will disburse funds on a no-regrets basis once three conditions are met:

- Endorsement by the Resident Coordinator of the framework (this document) and the CERF application package comprising an application strategy, agency-specific project proposals and agency-specific budgets;
- Pre-approval by the ERC of the framework (this document) and the CERF application package comprising an application strategy, agency-specific project proposals and agency-specific budgets (in annex); and
- Activation of the readiness trigger, the action trigger or the observational trigger within a maximum of two years from the ERC's pre-approval of the framework.

⁴ There is an additional \$4,000 per district from BDRCS as a decentralized fund for AA.

⁵ 50,000 GBP

As the CERF funding will be activated and distributed as automatically as possible immediately upon the defined triggers being reached, CERF will disburse funds on a no-regrets basis. While a portion of the funds can be used immediately upon disbursement for readiness activities, the remaining portion of the funding can only be used if and when an action or observational trigger is reached. The proportion of funding for readiness and for action activities will be clearly defined in the pre-approved project proposals and budgets. The need to distinguish between these two cost categories in advance is critical given the different potential scenarios.

Pre-arrangement and/or receipt of CERF funding for anticipatory action does not preclude or guarantee additional CERF funding for a complementary rapid response to save lives resulting from the same or other shocks. This means that the pre-arranged funding by CERF for anticipatory/early action is for the hazards and triggers outlined in this framework.

Challenges

- While financing is increasingly available to carry out anticipatory actions, collective anticipatory action requires an investment in building the framework. These “start-up costs” (or “build costs” as part of a “build – fuel – evaluate” cycle of frameworks) are incurred by participating agencies prior to the activation of a framework to set up effective anticipatory actions. These costs may also include projects which enable a collective approach (e.g. common targeting approaches) and other costs that pooled funds, including the CERF, traditionally do not fund.
- To enable a timely response within a short window of opportunity, anticipatory action at scale requires commensurate disaster risk preparedness, including the prepositioning of goods. There is limited funding available for such emergency response preparedness, thus limiting the scope of this anticipatory action framework.
- It must be understood that anticipatory action may support some of the most vulnerable communities facing the impact of a tropical cyclone. Yet, the storms are still likely to cause damage beyond the maximum number of people targeted, beyond the geographic reach or the scope of sectors covered by this anticipatory action framework. More thought needs to be invested in ensuring linkages between emergency response preparedness, anticipatory action, rapid response and early recovery, including around community engagement, activities and financing.

5. The Learning

Any activation of this framework will offer enhanced learning opportunities on anticipatory action. An ad hoc monitoring, evaluation and learning (MEAL) committee will be convened by OCHA with partners to this framework to coordinate learning and evidence gathering on anticipatory action.

One key aspect of learning will be to assess the impact of the framework against the premise of where possible, anticipatory actions may lead to a faster, more efficient, and more dignified humanitarian response, which also may protect development gains.

Learning from frameworks should be achieved at the highest possible standards and rigour. While the ad-hoc committee will discuss the key learning questions in more detail, the following areas of learning have been pre-identified as possible starting points designing monitoring and evaluation (M&E) as well as an independent evaluation:

- **Timing:** Did the timing of the intervention in relation to cyclone land-fall make a difference?
- **Joint targeting:** Did the common beneficiary database and joint targeting lead to better assistance provided to beneficiaries by agencies who used this modality?
- **Early warning:** How effective is the joint approach to early warning? Did the content of the messages matter?

- **Localisation:** Did the anticipatory / early action approach enable greater localization of the humanitarian response?
- **Partnership:** What can be learned from the partnership with the government, including how to support a take-up of anticipatory approaches?
- **Pre-crisis survey:** understand what are the risks of cyclone's impact on the community, households, and individuals that can be potentially mitigated prior to the hazard and therefore design effective Cyclone AA intervention
- **Actions:** How can the anticipatory action by each agency be further improved?
- **Effectiveness:** Was anticipatory or earlier action found effective in reducing the impact of the shocks?
- **AAP:** To what extent was the level of effective participation from communities ensured?

Monitoring & Evaluation

Each agency will use its existing M&E systems to collect and track data on implementation progress and outputs achieved. This can be financed through the CERF funds. All findings will be shared with OCHA. M&E questionnaires should be coordinated among agencies to allow for maximum learning, each to evaluate and report on a set of common questions across all interventions. This may also enable a reduction of interviews required by beneficiaries. The work will bring in MEAL experts at the global level and can draw on recently published guidance to evaluate anticipatory action.⁶ Further common questions can be discussed and agreed through the ad-hoc committee.

Independent Evaluation

Agencies participating in this framework also agree to participate in an independent evaluation, which may include the sharing of data on targeted people for this purpose. The ad hoc committee will oversee this evaluation and define the scope of the exercise.

6. Final provisions

Advocacy and Public Communications

Agencies agree to coordinate, under the leadership of the Resident Coordinator, key messages around the pilot, including during the development, activation and learning phase. In case of an activation of the pilot, close coordination facilitated by OCHA will support as a minimum: a joint press release and mutual amplification of social media stories.

To enhance common messaging and advocacy for anticipatory action, agencies agree to share any visuals (photos, videos) and impact stories with the ad hoc committee as these become available.

⁶ See for instance here: <https://www.anticipation-hub.org/news/new-guidance-on-planning-and-monitoring-country-capacity-strengthening-for-anticipatory-action>

7. Annexes

Annex 1 – Historical Analysis

Table 1: Historical data on cyclone impact on coastal areas of Bangladesh (used in return period calculation against wind speed, 118 km/h)

Year	Name	Wind speed (km/h)	Storm surge height (m)	Affected population	Affected houses (fully damaged)	Loss of life	Livestock
1991	Bob 1	250	6.5	15,000,000	854,399	138,882	700,000
2007	Sidr	223	4.57	6,737,815	564,967	> 10,000	382,000
2009	Aila	120	3.05	3,900,000	243,000	190	NA
2015	Komen	65	1-2	2.6 million	88,900	9	N/A
2016	Roanu	90	1.5	1,297,010	216,771	24	NA
2017	Mora	146	1.5	3,300,000	17,023	7	NA
2019	Fani	88 ⁷	1.5	1.6 million	2,363 ⁸	17	175
2019	Bulbul	140	1.83	2.1 million	16,580 ⁹	25	NA
2020	Amphan	151	3-5	2.6 million	55,667	26	NA
2021	Yaas	150 ¹⁰	1.8-2.4	1.3 million	26,000	9	NA
2022	Sitrang	88 ¹¹	1-2.4	1 million	20,000	35	NA
2023	Mocha	145	1.5-2	0.75 million	17,000	0	NA

⁷ https://rcbd-my.sharepoint.com/:b:/g/personal/md_salauddin_bdracs_org/ET-1jqDyZUVCguTx9iBEPFwBk9gdmd-rPqn20eSfvOKVmw?e=dKmzen

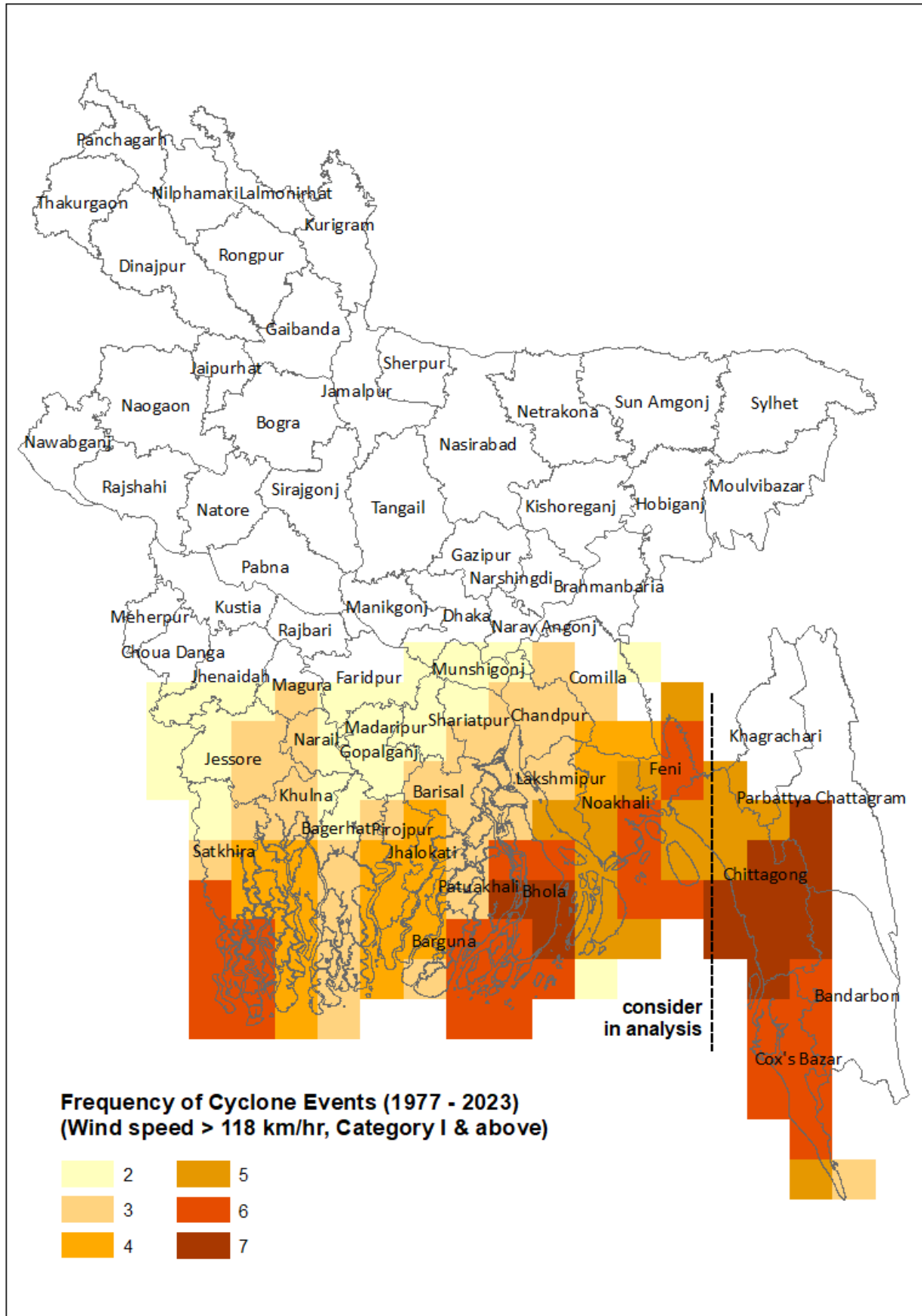
⁸ https://rcbd-my.sharepoint.com/:b:/g/personal/md_salauddin_bdracs_org/Ecnf7HelyIFCgqh1gc9iz8gBSv0XNTwpORcWSDSuQIGpPA?e=f7xE97

⁹ https://rcbd-my.sharepoint.com/:b:/g/personal/md_salauddin_bdracs_org/Ef1QmBTieOxKjNBe0AxlwSIBJ-QEHksa2YtjNOKGclz0w?e=5rRHUy

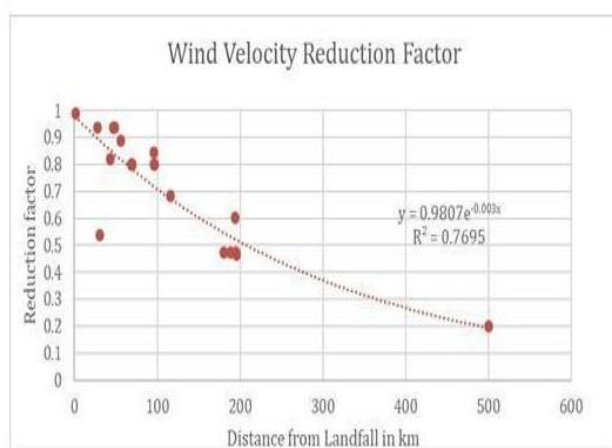
¹⁰ <https://reliefweb.int/report/bangladesh/bangladesh-cyclone-yaas-final-report-n-mdrbd027>

¹¹ <https://reliefweb.int/report/bangladesh/cyclone-sitrang-bangladesh-situation-report-2-26-october-2022#:~:text=Wind%20speed%20during%20landfall,%2D88%20kmph%2C%20during%20landfall>

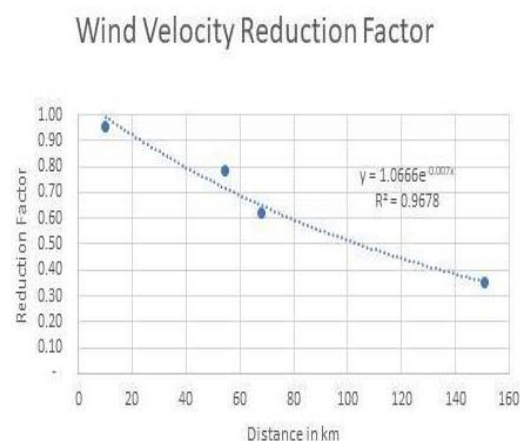
Map 1: Map of cyclone hit frequency (used in priority analysis for district selection to consider in pilot study)



Graph 1: Wind speed reduction factor model in respective to distance (sources: BMD & IMD)



Cyclone Amphan (2020)



Cyclone Mocha (2023)

Graph 2: HH asset damage curve with respect to wind speed differentiated by wall type

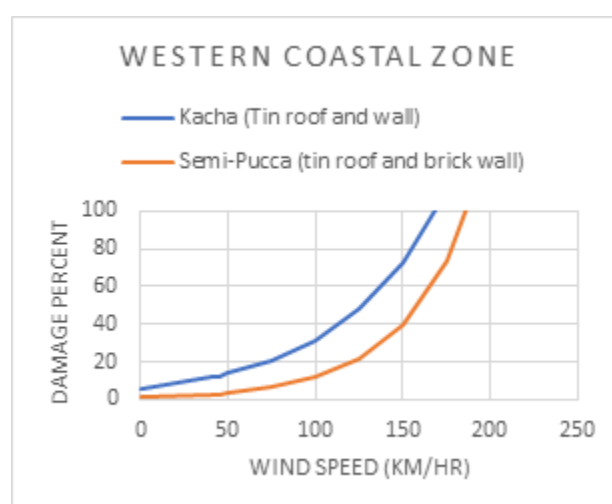
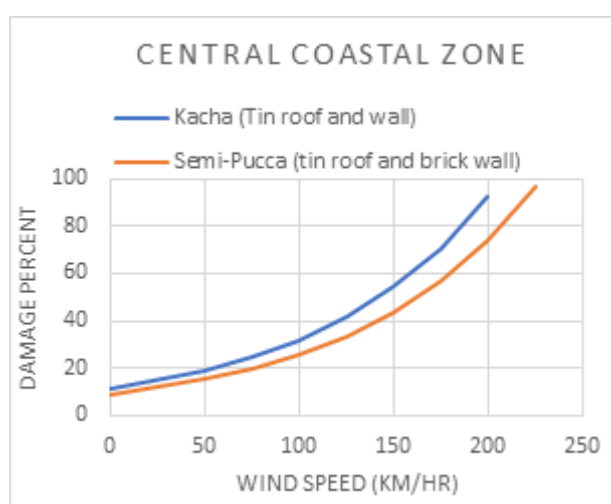


Table 2: Impacts (damage percentage) on rural houses for windspeed of 118 km/h

Zone	Semi Pucca (70%)	Kacha (30%)	Impact (weighted average)
Central	30	40	30
Western	20	45	25

Some 200 households were surveyed by BDRCS/GRC; structure failure modelling exercises were performed to analyse the impacts on rural house structures. This found that rural house structure damage due to 118 km/h winds in the central and western zone are respectively 30 per cent and 25 per cent. As such, it is assumed that wind speeds in excess of 118 km/h will cause structural damage to more than 25 percent of houses.

Table 3: Cyclone forecast models (reference are in the trigger statements)

Forecast	Content	Lead time	Skill
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BMD	Windspeed, Storm Surge (not location specific)	DD: 48 hr. CS: 30 hours	Wind velocity forecast skill is good when the storm is 1,000 km from the coast. The Storm Surge forecast needs improvement.
IMD	Track with wind speed and tentative storm surge	DD: 48-120 hr. SCS: 48 hr. +	After the formation of CS – the forecast is good, but storm surge improvement needed
ECMWF	Wind and storm surge	10 days	After the formation of DD, skill is good
GFS	Wind and storm surge	10 days	After formation of D/DD, skill is good
JTWC	Track with wind speed and tentative storm surge	After the formation of DD	After the formation of CS – the forecast is good

Source: EAP Cyclone, IFRC

Performance of forecast models

The draft national cyclone EAP for Bangladesh shows the forecast errors (table below) of the model based on the reanalysis information of RSMC and publications.

Organization/ Model	Parameter	Lead time (ahead of time)		
		24 hour	48 hour	72 hour
BMD	Average Track Forecast Error (km)	60-80	80-100	100-120
	Average Intensity (wind) forecast Error (km/hr)	15-20	20-30	20-40
	Average Landfall Time forecast errors (hour)	1-2	3-4	4-6
RSMC-IMD	Average Track Forecast Error (km)	65	90	115
	Average Intensity (wind) forecast Error (km/hr)	17	20	20
	Average Landfall Time forecast errors (hour)	1.8	2.7	4.7
JTWC	Track Forecast Error (km)	24	54	80
	Intensity (wind) forecast Error (km/hr)	13.7	12.9	14.1
	Landfall time forecast errors (hour)	3	5	6
ECMWF Model	Track Forecast Error (km)	78	103	154
	Intensity (wind) forecast Error (km/hr)	25	40	45
	Landfall time forecast errors (hour)	2.3	2.9	4.2
NCEP-GFS Model	Track Forecast Error (km)	78	115	170
	Intensity (wind) forecast Error (km/hr)	25	35	45
	Landfall time forecast errors (hour)	2.2	3.9	6.7

Above table shows, the model track forecast beyond 48 hours the error is very high. Hence can be used only for readiness but not for activation. The severity of impact on livelihood is high when a cyclonic storm becomes a Very Severe Cyclonic Storm or more and needs support from the government.

The monitoring team is composed of Ahmadul Hassan (a research assistant), Pauline Ndirangu (Data Scientist, OCHA) and Leonardo Milano (Data Science team Lead, OCHA).

Annex 2 – Distribution list

To	Gwyn Lewis, Resident coordinator Office (lewisg@un.org)
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Annex 3 – Trigger Messages

From:

To: distribution list (see annex)

[readiness trigger]

SUBJECT: Cyclone Anticipatory Action Bangladesh – Readiness Trigger Reached

Dear Resident Coordinator, dear colleagues,

The readiness trigger of the coordinated anticipatory action pilot for tropical cyclones in Bangladesh has been reached in line with the forecasts shared by the Bangladesh Meteorological Department (BMD), India Meteorological Department (IMD), European Centre for Medium-Range Weather Forecasts (ECMWF), Global Forecast System (GFS) and Joint Typhoon Warning Center (JTWC). The forecasts indicate that wind speeds in the unions of the coastal districts are expected to exceed 118 km/h.

Readiness activities in line with the [framework document](#) should now commence. CERF funding to UN agencies is being released. OCHA may convene a meeting of the participating partners.

Best wishes and good luck,

[action trigger]

SUBJECT: Cyclone Anticipatory Action Bangladesh – Activation Trigger Reached

Dear Resident Coordinator, dear colleagues,

The activation trigger of the coordinated anticipatory action pilot for tropical cyclones in Bangladesh has been reached in line with the official forecasts shared by the Bangladesh Meteorological Department (BMD), India Meteorological Department (IMD), European Centre for Medium-Range Weather Forecasts (ECMWF), Global Forecast System (GFS) and Joint Typhoon Warning Center (JTWC). The forecasts indicate that wind speeds in the unions of the coastal districts are expected to exceed 118 km/h.

All activities in line with the [framework document](#) should now commence. OCHA may convene a meeting of the participating partners.

Thank you all for the dedication and expertise you have brought to the framework development. Your efforts have been instrumental.

Best wishes and good luck,

PS: Please remember that the impact of this pilot will be evaluated. We encourage all partners to document decisions and lessons learned as best as possible during this activation. Finally, please also ensure any visuals, stories or social media posts are shared with the group.

[observational trigger]

SUBJECT: Cyclone Anticipatory Action Bangladesh – Observational Trigger Reached

Dear Resident Coordinator, dear colleagues,

The observational trigger of the coordinated anticipatory action pilot for tropical cyclones in Bangladesh has been reached in line with the observed data by the Bangladesh Meteorological Department (BMD), India Meteorological Department (IMD), European Centre for Medium-Range Weather Forecasts (ECMWF), Global Forecast System (GFS) and Joint Typhoon Warning Center (JTWC). The observed data indicate that wind speeds at landfall in the unions of the coastal districts exceeded 118km/h.

All activities in line with the [framework document](#) should now commence. OCHA may convene a meeting of the participating partners.

Thank you all for the dedication and expertise you have brought to the framework development. Your efforts have been instrumental.

Best wishes and good luck,

PS: Please remember that the impact of this pilot will be evaluated. We encourage all partners to document decisions and lessons learned as best as possible during this activation. Finally, please also ensure any visuals, stories or social media posts are shared with the group.

[map of track and table of unions after pre-activation or activation trigger is reached]

SUBJECT: Cyclone Anticipatory Action Bangladesh – Forecast Information

Dear colleagues,

The forecast for cyclone NAME produced at HH:MM on YYYY-MM-DD (BDT) indicates that wind speeds in the unions of the coastal districts are expected to exceed 118 km/h.

A plot of the forecasted track is shown below.

[Image of the track from BMD]

The following table displays unions with wind speeds exceeding 118 km/h for unions in Barguna, Bhola, Patuakhali, Noakhali, and Satkhira.

[Table with unions from the monitoring spreadsheet]

Please refer to the activation criteria outlined in the [Anticipatory Action Framework: Bangladesh Cyclones](#) for further details.

Thank you all for the dedication and expertise you have brought to the framework development. Your efforts have been instrumental.

Best wishes.

Annex 4 – Overview of Activities

Agency	Sector/Cluster	CERF Funds (US\$)	Co-fund (US\$)	People Targeted	Key Activities
FAO	Food Security - Agric	\$500,000		106,500	Dissemination of early warning advisories to remote farming communities (25,000 HH)
					Shelter based cattle management and cattle feed distribution (12,000 people at 60 shelter centers)
					Information, Education and Communication materials on livestock management in emergencies
					Unconditional cash support to protect farming livelihoods (23,643 people)
WFP	Multi-purpose cash	\$2,100,000	\$ 1,600,000	329,000	Multipurpose Cash Support, Early warning
UNICEF	WASH	\$780,000	1,900,000	150,000	Deploy 20 Mobile Water Treatment Plants in the potential evacuation point and Cyclone Shelters to provide safe drinking water
					Distribution of life-saving WASH supplies: Water Purification Tablets (WPTs), Jerrycans, Soap and Sanitary Napkin
					Minor repair/ maintenance /installation of temporary WASH facilities (water supply, sanitation and hand washing) at evacuation point and cyclone shelters
					Mass communication on early warning and hygiene promotion (handwashing, use of latrines, use of WPT, Menstrual hygiene) using media, banners and posters in cyclone shelters for better visibility.

	Health	\$220,000		61,191	Anticipatory / early action for cyclones through deployment of mobile medical team to deliver quality essential services to cyclone impacted communities with lifesaving supplies, medicines; provide transport services for critically ill patients; refer pregnant mothers to higher level facilities; prepositioning of essential medications; provide ANC and safe delivery services.
	Multi-purpose cash	\$99,000		1,940	Providing unconditional multipurpose cash support to cyclone-affected vulnerable households targeting pregnant and lactating mothers with young children (ages 0-4 years)
UNFPA	Protection - GBV	\$88,235	\$15,930	2,500	- Distribution 2500 Dignity Kit along with lifesaving GBV risk mitigation information and referral pathways to women and other vulnerable groups - Deployment of 40 trained Women and Girls Support Volunteers for DK distribution, dissemination of safeguarding messages including early warning messages, and GBV referral pathway. Volunteers will be specifically trained to support women with disabilities and their caregivers. - Deployment of 10 MHPSS counselors in evacuation centres during the cyclone to provide PFA and lifesaving information, addressing heightened protection risks for vulnerable women and girls
	Health/ - SRH	\$211,765		6,000	- Deployment of 30 midwives, 08 medical doctors, and 60 volunteers in temporary safe evacuation centres for pregnant women. - Distribution of 2,000 baby kits to pregnant women with transportation cash support - Distribution of 3,000 MHM kits including transportation cost among adolescent girls including girls with disabilities - Distribution of cash for Menstrual Health and Hygiene to 1,000 adolescent girls including girls with disabilities - Deployment of trained 170 youth volunteers at the evacuation centers and distribution points to facilitate distribution, safeguarding message dissemination including early warning messages, and GBV referral pathway. Volunteers will be specifically trained to support girls with disabilities and their caregivers.